

TEAM UPDATE: ISC2 x CYBLAC Cybersecurity Bootcamp Team Update 4 (June 28 - July 5, 2025)

Hello Team,

Network Security was led with exceptional expertise by Nkeiruka Madu, last week's sessions equipped us with a comprehensive understanding of how networks operate, the vulnerabilities they face, and the strategies to protect them. Below is a detailed recap of the sessions, highlighting key events, essential network security concepts, and actionable next steps to keep us on track.

Key Events and Session Highlights

Saturday, June 28: Introduction to Network and Network Security

Event: Nkeiruka Madu kicked off the three-day network security training. The session was thoughtfully divided into two segments, with dedicated Q&A time to address our questions.

Key Topics:

- **Network Fundamentals and Architecture:** Hubs, switches, routers, firewalls, servers, endpoints, and Ethernet cables were explained to network enabling resource sharing and significantly reducing operational costs. We explored the four primary network types: Personal Area Network (PAN), Local Area Network (LAN), Metropolitan Area Network (MAN), and Wide Area Network (WAN). Data transmission was unpacked through the OSI model's seven layers and the more streamlined TCP/IP architecture, which condenses these into four layers for practical implementation.
- **Network Addressing and Protocols:** A key focus was on addressing mechanisms, starting with the MAC addresses for identifying physical networking devices and IP addresses (IPv4 and IPv6) as logical identifiers. We looked at TCP's reliable three-way handshake and contrasted by UDP's speed for low-latency needs.
- **Wireless Networking:** Wi-Fi was presented as a wireless technology enabling cable-free connections, praised for its easy setup, flexibility, and portability. However, its vulnerability to attackers was a critical takeaway, mitigated by using strong

passwords and limiting network access.

Sunday, June 29: Network and Network Security Continued

Event: Nkeiruka Madu built on the previous day, deepening our understanding.

Key Topics:

- **Cloud Computing:** We explored cloud computing's origins, benefits, and deployment models. We learned that cloud services are delivered over the internet via a pay-as-you-go model, offering on-demand provisioning and flexible usage. The four deployment types were public, private, hybrid, and community clouds. The three primary service models (IaaS, PaaS, and SaaS) were explained. Cloud Service Providers (CSPs), Managed Service Providers (MSPs) and Service Level Agreements (SLAs) were other terms that were discussed.
- **Cybersecurity Threats and Detection:** We explored how network communications can be intercepted via man-in-the-middle attacks, even with encryption. Threats were categorized as natural (disasters), accidental (human error), or human-made (insider threats), all compromising the CIA triad. Specific threats like phishing (via emails, SMS, calls), viruses, worms, Trojan horses, ransomware, and APTs were detailed as well. Security monitoring systems were introduced, including Intrusion Detection Systems (IDS), Security Information and Event Management (SIEM) and Intrusion Prevention Systems (IPS) to block threats.
- **Network Security Best Practices:** Recommended practices included enabling firewalls, keeping systems updated, using IDS/IPS, removing unused services, and maintaining updated antivirus (noting expired antivirus is more dangerous than none).

Saturday, July 5: Network Segmentation and Security Strategies

Event: Nkeiruka Madu concluded the week with advanced security strategies.

Key Topics:

- **Data Center Infrastructure:** Traditional data centers remain vital despite cloud growth, comprising servers, networking equipment, cabling, and environmental controls. Maintaining HVAC systems prevents temperature damage, while gas-based fire suppressants protect electrical environments. Redundancy across servers, resources, and power supplies (with backup generators and UPS) prevents single points of failure. Hot, warm, and cold recovery sites were discussed for disaster recovery, alongside seamless power changeovers and redundant environments to ensure business continuity.
- **Memorandums of Understanding (MOUs)** enable organizations to share recovery resources during outages, even among competitors, provided both have hot or warm sites, fostering collaboration on safety and threat information.
- **Network Segmentation:** Segmentation methods include DMZ, VLANs, MAC addresses, IP addresses, physical ports, protocols, and applications. Zero trust security requires continuous authentication and authorization, complemented by the principle of least privilege, granting only minimal permissions.

Action Items and Reminders

- *Tasks (Due July 25, 2025):*
 1. Network Security Task is available on ClickUp, requiring detailed analysis and submission.
 2. Security Operations task is yet to be updated.
- **Next Session:** Saturday, July 6, 2025, with the next session Saturday, July 12, 2025 (TODAY) followed by the Explanation on Sunday, July 13, 2025.

Best regards,

Prince Kwesi Lassey (Team Lead) & Joachim Araga (Assistant Team Lead)